

# Notes: Amihud and Noh (RFS, 2021)

## Illiquidity and Stock Returns II: Cross-section and Time-series Effects

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# 1 Big picture

- **Research question.** Does Amihud’s illiquidity measure,  $ILLIQ$ , contain priced illiquidity information beyond its inverse-dollar-volume component,  $IDVOL$ ? More broadly, is the return premium associated with  $ILLIQ$  an illiquidity premium or merely a volume, mispricing, sentiment, or seasonality effect?
- **Main debate.** Lou and Shu argue that the average inverse dollar volume,  $IDVOL$ , is sufficient to explain why  $ILLIQ$  predicts returns. Amihud and Noh argue that this conclusion follows from an incomplete decomposition of  $ILLIQ$ .
- **Core idea.**  $ILLIQ$  is the average of a product,  $|r|/dvol$ . The average of a product is not generally equal to the product of averages. The missing covariance term contains economically meaningful illiquidity information.
- **Key variable.** The paper defines

$$DIF_{j,s} = \ln ILLIQ_{j,s} - \ln LSilliq_{j,s} = \ln ILLIQ_{j,s} - [\ln |R_{j,s}| + \ln IDVOL_{j,s}].$$

$DIF$  captures the information in  $ILLIQ$  that is lost when one replaces  $ILLIQ$  with the Lou-Shu product of averages.

- **Main cross-sectional finding.**  $DIF$  is positively and significantly priced in Fama-MacBeth regressions. The residual component of  $\ln ILLIQ$  orthogonal to  $\ln IDVOL$  is also positively priced.
- **Main time-series finding.** Shocks to aggregate market  $ILLIQ$  explain realized market returns much better than shocks to aggregate  $IDVOL$ . The coefficient on market  $ILLIQ$  shocks is negative, as theory predicts: higher illiquidity raises expected returns and lowers contemporaneous prices.
- **Robustness.** The illiquidity premium remains positive after controlling for mispricing, sentiment, January seasonality, idiosyncratic volatility, and systematic risk loadings.
- **Economic takeaway.** Trading volume is informative about liquidity, but it is not a sufficient substitute for  $ILLIQ$ . The return-impact component of  $ILLIQ$  carries additional information about illiquidity and is priced in both the cross-section and the time-series.

# 2 Incremental contribution of the paper

- **Corrects the Lou-Shu decomposition.** The paper shows that Lou and Shu’s decomposition effectively assumes  $\text{cov}(|r|, 1/dvol) = 0$ . Because this covariance is not zero, their decomposition omits a term.
- **Defines and tests the omitted component.**  $DIF$  measures the gap between  $\ln ILLIQ$  and  $\ln LSilliq$ . The paper shows that this omitted component has strong explanatory power for expected returns.

- **Connects the omitted component to microstructure theory.** The missing term is approximated using the slope  $b$  from a regression of  $|r|$  on dollar volume and the coefficient of variation of dollar volume,  $CV$ . The slope  $b$  is related to price impact and therefore to illiquidity.
- **Goes beyond cross-sectional pricing.** The paper also studies aggregate market illiquidity shocks and shows that market *ILLIQ* is superior to market *IDVOL* in explaining contemporaneous realized returns.
- **Addresses alternative explanations.** The paper directly tests whether the illiquidity premium is a January effect, a mispricing effect, or a sentiment effect. It finds that the premium survives these controls.
- **Uses microstructure benchmarks.** The residual component of  $\ln ILLIQ$  is positively related to Kyle's  $\lambda$  and quoted bid-ask spreads, supporting the interpretation that it captures illiquidity rather than only trading activity.
- **Clarifies the role of volume.** The paper does not deny that volume is related to liquidity. Instead, it argues that volume alone leaves out part of the illiquidity signal in *ILLIQ*.

## 3 Data and measurement

### 3.1 Stock-level sample

The main sample is a monthly panel of U.S. common stocks over January 1955 to December 2016, for 744 months.

- **Stocks.** The sample includes NYSE and AMEX common stocks with CRSP share codes 10 or 11.
- **Price filter.** Stocks are required to have an average price between \$5 and \$1000 over the 12-month estimation window.
- **Daily data filters.** Stock-days with negative prices, trading volume below 100 shares, or returns below  $-100\%$  are removed.
- **Window.** Illiquidity variables are computed using daily return and volume data over a 12-month period ending in month  $s$ .
- **Data requirement.** The 12-month window must contain at least 200 trading days.
- **Outlier treatment.** For each stock-month, the day with the highest value of *ILLIQ*, *LSIlliq*,  $|R|$ , or *IDVOL* is excluded. Extreme 1% stock-months are also censored for several variables.

### 3.2 Illiquidity variables

For stock  $j$  on day  $d$  in month-window  $s$ , daily Amihud illiquidity is

$$illiq_{j,d,s} = \frac{|r_{j,d,s}|}{dvol_{j,d,s}},$$

where  $dvol$  is dollar trading volume in millions.

The main stock-level variables are:

- $ILLIQ_{j,s}$ : the average daily value of  $|r|/dvol$ .
- $|R_{j,s}|$ : the average daily absolute return.
- $IDVOL_{j,s}$ : the average daily inverse dollar volume,  $1/dvol$ .
- $LSIlliq_{j,s} = |R_{j,s}| \times IDVOL_{j,s}$ : the Lou-Shu product-of-averages component.
- $DIF_{j,s} = \ln ILLIQ_{j,s} - [\ln |R_{j,s}| + \ln IDVOL_{j,s}]$ : the missing log component.
- $RlnILLIQ_{j,s}$ : the residual from a monthly cross-stock regression of  $\ln ILLIQ_{j,s}$  on  $\ln IDVOL_{j,s}$ .

### 3.3 Control variables

The baseline Fama-MacBeth regressions control for standard stock characteristics:

- *Size*: log market capitalization.
- *BM*: log book-to-market ratio.
- *R1lag*: return over the previous month, controlling for short-term reversal.
- *R12lag*: cumulative return over months  $s - 2$  through  $s - 12$ , controlling for momentum.
- *IdioVol*: idiosyncratic volatility from daily residuals after controlling for Fama-French three factors.
- Systematic risk loadings on *RMrf*, *SMB*, *HML*, and *UMD* in robustness tests.
- *MISP*: the Stambaugh-Yu-Yuan mispricing rank based on 11 anomaly variables.

### 3.4 Aggregate market illiquidity variables

For each month  $t$ , the paper constructs market-wide illiquidity series:

- $mILLIQ_t$ : the logarithm of the value-weighted cross-stock average of monthly  $ILLIQ$ .
- $mIDVOL_t$ : the logarithm of the value-weighted cross-stock average of monthly  $IDVOL$ .
- $m|R|_t$ : the logarithm of the value-weighted cross-stock average of monthly absolute returns.

- $dmILLIQ_t$ ,  $dmIDVOL_t$ , and  $dm|R|_t$ : shocks to these market series, computed as out-of-sample prediction errors from rolling AR(2) models using the prior 60 months.
- $dmDIF_t = dmILLIQ_t - (dm|R|_t + dmIDVOL_t)$ .

Interpretation. The cross-sectional variables measure whether more illiquid stocks earn higher expected returns. The time-series variables measure whether shocks to market illiquidity move aggregate realized returns in the direction predicted by liquidity-pricing theory.

## 4 Methodology and empirical specifications

### 4.1 Decomposition of $ILLIQ$

Amihud's daily illiquidity measure is averaged over a period:

$$ILLIQ = \overline{illiq_d} = \frac{\overline{|r_d|}}{dvol_d}.$$

The expected value of the daily product is

$$E\left(\frac{|r_d|}{dvol_d}\right) = E(|r_d|)E\left(\frac{1}{dvol_d}\right) + \text{cov}\left(|r_d|, \frac{1}{dvol_d}\right).$$

Lou and Shu's decomposition uses

$$LSIlliq = \overline{|r_d|} \times \frac{1}{\overline{dvol_d}},$$

which ignores the covariance term. Thus,

$$ILLIQ = LSIlliq + \text{cov}\left(|r_d|, \frac{1}{dvol_d}\right).$$

Using a Taylor-series approximation,

$$\text{cov}\left(|r_d|, \frac{1}{dvol_d}\right) \approx -b \times CV^2,$$

where  $b$  is the slope from a regression of  $|r_d|$  on  $dvol_d$  and  $CV$  is the coefficient of variation of dollar volume. Therefore,

$$ILLIQ \approx LSIlliq - bCV^2,$$

and

$$DIF = \ln ILLIQ - \ln LSIlliq \approx \ln\left(1 - \frac{bCV^2}{LSIlliq}\right).$$

Interpretation. The omitted component is not a mechanical nuisance. It is related to the way price movements co-vary with trading activity, which is central to the economic meaning of illiquidity.

## 4.2 Fama-MacBeth cross-sectional regressions

The baseline cross-sectional specification is estimated monthly:

$$(R_j - r_f)_s = b_{0s} + b'_{1s} IL_{j,s-2} + b_{2s} Size_{j,s-2} + b_{3s} BM_{j,y-1} + b_{4s} R12lag_{j,s-2} + b_{5s} R1lag_{j,s-1} + \varepsilon_{j,s}.$$

The vector  $IL_{j,s-2}$  alternately includes  $\ln ILLIQ$ ,  $\ln LSilliq$ ,  $\ln |R|$ ,  $\ln IDVOL$ ,  $DIF$ , or  $RlnILLIQ$ . The two-month lag follows the earlier literature and reduces timing concerns.

Interpretation. A positive coefficient on an illiquidity variable means that stocks that are more illiquid in the recent past earn higher subsequent expected returns.

## 4.3 Horse-race residual specification

Because  $\ln ILLIQ$  and  $\ln IDVOL$  are highly correlated across stocks, a simple horse race can be unstable. The paper therefore estimates, each month,

$$\ln ILLIQ_{j,s} = a_s + c_s \ln IDVOL_{j,s} + RlnILLIQ_{j,s}.$$

Then it tests whether  $RlnILLIQ_{j,s-2}$  predicts returns in the Fama-MacBeth regression while also including  $\ln IDVOL_{j,s-2}$ .

Interpretation. If  $RlnILLIQ$  is priced, then  $ILLIQ$  contains priced information beyond inverse dollar volume.

## 4.4 Mispricing, sentiment, and seasonality tests

The paper addresses three alternative explanations:

- **Mispricing.** Add the Stambaugh-Yu-Yuan mispricing rank to the cross-sectional regressions and also test whether illiquidity premia remain after controlling for mispricing factors.
- **Seasonality.** Regress monthly illiquidity premia on a January dummy and market returns to test whether the premium disappears outside January.
- **Sentiment.** Test whether lagged Baker-Wurgler sentiment explains the illiquidity premium, and whether sentiment changes explain the effect of market illiquidity shocks on aggregate returns.

## 4.5 Time-series regressions of market illiquidity shocks

The time-series model is

$$RMrf_t = a + b'dY_t + \varepsilon_t,$$

where  $dY_t$  can include  $dmILLIQ_t$ ,  $dmIDVOL_t$ ,  $dm|R|_t$ , and  $dmDIF_t$ .

The paper also estimates analogous models with  $SMB_t$  as the dependent variable and includes  $RMrf_t$  as a control.

Interpretation. Since market illiquidity is persistent, a positive shock to illiquidity raises required returns and should cause a contemporaneous price decline. Therefore, the expected coefficient on  $dmILLIQ_t$  is negative.

## 4.6 Benchmark and crisis tests

To assess which market series better captures illiquidity, the paper compares  $mILLIQ$  and  $mIDVOL$  with benchmark illiquidity measures: Kyle's  $\lambda$ , quoted bid-ask spreads, effective spreads, and Pastor-Stambaugh liquidity innovations. It also examines the behavior of  $mILLIQ$  and  $mIDVOL$  around the October 1987 crash and the 2008 financial crisis.

# 5 Identification and interpretation

## 5.1 What the paper can identify

The paper identifies whether the information in Amihud's  $ILLIQ$  beyond inverse dollar volume is priced and whether market-wide  $ILLIQ$  shocks better explain realized aggregate returns than market  $IDVOL$  shocks. The evidence is reduced-form asset-pricing evidence, not a structural estimate of trading costs.

## 5.2 Why the decomposition matters

The key econometric point is simple but important:  $ILLIQ$  is an average of a ratio, while Lou and Shu's measure is a product of two averages. These are the same only under a strong covariance restriction. The omitted covariance term is exactly where Amihud and Noh find additional illiquidity information.

## 5.3 Why $DIF$ is interpreted as illiquidity-related

The approximation  $DIF \approx \ln(1 - bCV^2/LSIlliq)$  links  $DIF$  to two objects with liquidity meaning:

- $b$ , the slope of absolute returns on dollar volume, resembles a price-impact relation.
- $CV^2$ , variation in trading volume, affects the timing of trading costs and expected returns.

The paper also shows that the residual component of  $\ln ILLIQ$  is positively related to Kyle's  $\lambda$  and quoted bid-ask spreads, strengthening the interpretation that it captures illiquidity rather than only volume.



## 5.4 Cross-sectional interpretation

The Fama-MacBeth coefficients are return premia in percent. The positive and significant coefficients on  $DIF$  and  $RlnILLIQ$  imply that the part of  $ILLIQ$  not explained by  $IDVOL$  is rewarded in expected returns.

## 5.5 Time-series interpretation

A negative coefficient on  $dmILLIQ_t$  means that when market illiquidity unexpectedly rises, realized market returns fall. This is consistent with the standard asset-pricing logic that higher future required returns imply lower current prices.

## 5.6 Alternative explanations

The paper’s controls for mispricing, sentiment, and January seasonality are important because the volume and illiquidity premia could otherwise be interpreted as behavioral anomalies. The results indicate that the illiquidity premium is not fully explained by these alternatives.

## 5.7 Limitations

$ILLIQ$  is a proxy, not a complete measure of all dimensions of liquidity. It blends price impact, fixed trading costs, volatility, and trading activity. The paper shows that  $ILLIQ$  is useful and contains additional priced information, but it does not claim that  $ILLIQ$  is the unique or exhaustive measure of illiquidity.

# 6 Tables and figures

## 6.1 Table 1: Cross-sectional pricing of $ILLIQ$ and its components

Read:

- Table 1 reports Fama-MacBeth cross-sectional regressions for January 1955 to December 2016.
- $\ln ILLIQ$  is positively priced: the coefficient is 0.102 with a  $t$ -statistic of 2.45.
- $DIF$  is positively and strongly priced. Its coefficient is 1.219 when controlling for  $\ln LSilliq$ , and 0.996 when controlling for  $\ln IDVOL$  and  $\ln |R|$ .
- $RlnILLIQ$ , the residual part of  $\ln ILLIQ$  orthogonal to  $\ln IDVOL$ , is also positively priced with a coefficient of 0.736 and a  $t$ -statistic of 4.63.

- *IDVOL* remains positively priced, so the point is not that volume is irrelevant. The point is that volume is incomplete.

Economic message: The omitted component in *ILLIQ* is not noise. It carries priced information about illiquidity beyond inverse dollar volume.

## 6.2 Table 2: Market illiquidity shocks and realized returns

Read:

- Table 2 estimates time-series regressions of realized stock returns on shocks to aggregate market illiquidity series.
- $dmILLIQ_t$  has a strongly negative effect on  $RMrf_t$ : the coefficient is  $-0.122$  with a  $t$ -statistic of  $-12.35$ .
- $dmIDVOL_t$  alone is also negative, but its explanatory power is much weaker.
- When  $dmILLIQ_t$  and  $dmIDVOL_t$  are included together,  $dmILLIQ_t$  remains negative and significant while  $dmIDVOL_t$  becomes positive in the market-return regression.
- $dmDIF_t$  is negative and significant when the components are included together, matching the positive cross-sectional pricing of *DIF*.
- Results for  $SMB_t$  show that illiquidity shocks affect small stocks more strongly, as expected.

Economic message: Market *ILLIQ* is a better time-series proxy for aggregate illiquidity shocks than market *IDVOL*. The sign of  $dmILLIQ$  is consistent with liquidity-pricing theory.

**Table 1**  
The effect of illiquidity variables on expected stock return: Fama-Macbeth cross-sectional regressions

| Explanatory variables                                                                 | (1)          | (2)                 | (3)                 | (4)                 |
|---------------------------------------------------------------------------------------|--------------|---------------------|---------------------|---------------------|
| $\ln ILLIQ_{j,s-2}$                                                                   | 0.102 (2.45) |                     |                     |                     |
| $\ln SILLIQ_{j,s-2}$                                                                  |              | 0.111 (2.64)        |                     |                     |
| $\ln R_{j,s-2}$                                                                       |              |                     | -0.317 (-2.10)      |                     |
| $\ln IDVOL_{j,s-2}$                                                                   |              |                     | 0.093 (2.64)        | 0.099 (3.01)        |
| $DIF_{j,s-2}$                                                                         |              | <b>1.219 (4.42)</b> | <b>0.996 (3.75)</b> |                     |
| $RlnILLIQ_{j,s-2}$                                                                    |              |                     |                     | <b>0.736 (4.63)</b> |
| $IdioVol_{j,s-2}$                                                                     |              |                     |                     | -0.663 (-8.1)       |
| Control variables: $Size_{j,s-2}$ , $BM_{j,s-1}$ , $R12lag_{j,s-2}$ , $RIlag_{j,s-1}$ |              |                     |                     |                     |
| Average Adjusted $R^2$                                                                | 5.62%        | 5.95%               | 7.48%               | 7.52%               |

This table presents the averages of slope coefficients from monthly Fama-Macbeth cross-sectional regressions

Table 1: Fama-MacBeth pricing of *ILLIQ*, *IDVOL*, and *DIF*.

**Table 2**  
The effect of shocks in market illiquidity series on realized stock returns

|                   | A. $RMrf_t$        |                   |                   |                                 | B. $SMB_t$        |                                 |
|-------------------|--------------------|-------------------|-------------------|---------------------------------|-------------------|---------------------------------|
|                   | (1)                | (2)               | (3)               | (4)                             | (5)               | (6)                             |
| $dmILLIQ_t$       | -0.122<br>(-12.35) |                   | -0.156<br>(-9.57) |                                 | -0.063<br>(-6.66) |                                 |
| $dmIDVOL_t$       |                    | -0.061<br>(-5.48) | 0.051<br>(3.21)   | -0.103<br>(-10.94)              | 0.014<br>(1.61)   | -0.047<br>(-6.83)               |
| $dmR_{I,t}$       |                    |                   |                   | -0.155<br>(-9.72)               |                   | -0.063<br>(-6.70)               |
| $dmDIF_t$         |                    |                   |                   | <b>-0.143</b><br><b>(-6.32)</b> |                   | <b>-0.051</b><br><b>(-3.86)</b> |
| $dm\lambda_{I,t}$ |                    |                   |                   |                                 |                   |                                 |
| $RMrf_t$          |                    |                   |                   |                                 | 0.079<br>(2.47)   | 0.078<br>(2.42)                 |
| Adjusted $R^2$    | 0.23               | 0.05              | 0.25              | 0.25                            | 0.15              | 0.15                            |

Table 2, panels A-B: Market illiquidity shocks and  $RMrf_t$  /  $SMB_t$ .

## Appendix C: Additional Empirical Analyses

Table C.1  
Summary statistics of the *ILLIQ*-related variables

| Variables           | Mean  | Standard<br>Deviation | Pairwise Correlations |                     |                 |                   |
|---------------------|-------|-----------------------|-----------------------|---------------------|-----------------|-------------------|
|                     |       |                       | $\ln ILLIQ_{j,s}$     | $\ln LSilliq_{j,s}$ | $\ln  R_{j,s} $ | $\ln IDVOL_{j,s}$ |
| $\ln ILLIQ_{j,s}$   | -3.34 | 1.98                  | 1.0                   |                     |                 |                   |
| $\ln LSilliq_{j,s}$ | -3.17 | 2.02                  | 0.99                  |                     |                 |                   |
| $\ln  R_{j,s} $     | -4.13 | 0.34                  | 0.45                  |                     |                 |                   |
| $\ln IDVOL_{j,s}$   | 0.97  | 1.89                  | 0.99                  | 0.99                | 0.30            |                   |
| $DIF_{j,s}$         | -0.17 | 0.09                  | -0.37                 | -0.41               | -0.24           | -0.38             |

For each stock  $j$ , we calculate the averages of the daily values of  $illiq_{j,d,s} = |r_{j,d,s}| / dvol_{j,d,s}$ ,  $|r_{j,d,s}|$ , and  $1/dvol_{j,d,s}$ , where  $r_{j,d,s}$  and  $dvol_{j,d,s}$  are, respectively, the daily return and dollar trading volume on day  $d$ . The averages of these values for each stock over the preceding 12 months that end in month  $s$  are  $ILLIQ_{j,s}$ ,  $|R_{j,s}|$ , and  $IDVOL_{j,s}$ , respectively. We also define  $LSilliq_{j,s} = |R_{j,s}| * IDVOL_{j,s}$ . The prefix "ln" indicates natural logarithm.  $DIF_{j,s} = \ln ILLIQ_{j,s} - \ln LSilliq_{j,s} = \ln ILLIQ_{j,s} - [\ln |R_{j,s}| + \ln IDVOL_{j,s}]$ . The table presents the time-series averages of the monthly cross-sectional statistics of the variables over the sample period of 1955 to 2016, 744 months.

|                 | C. $RMrf_t$       |                   |                   |                                 | D. $SMB_t$        |                   |
|-----------------|-------------------|-------------------|-------------------|---------------------------------|-------------------|-------------------|
|                 | (7)               | (8)               | (9)               | (10)                            | (11)              | (12)              |
| $dmILLIQ_t$     |                   | -0.084<br>(-4.22) |                   |                                 | -0.057<br>(-4.76) |                   |
| $dmDVOI_t$      |                   |                   | -0.013<br>(-0.69) | -0.055<br>(-2.97)               |                   | -0.017<br>(-1.29) |
| $dm R_t $       |                   |                   |                   | -0.129<br>(-4.99)               |                   |                   |
| $dmDIF_t$       |                   |                   |                   | <b>-0.103</b><br><b>(-3.21)</b> |                   |                   |
| $dm\lambda_t$   | -0.843<br>(-3.23) | -1.790<br>(-1.39) | -4.709<br>(-3.12) | -0.548<br>(-0.42)               | -0.127<br>(-0.12) | -1.717<br>(-1.73) |
| $RMrf_t$        |                   |                   |                   |                                 | 0.058<br>(1.11)   | 0.124<br>(2.54)   |
| <i>Adjusted</i> | 0.04              | 0.14              | 0.04              | 0.18                            | 0.11              | 0.05              |

Table 2, panels C-D: Controls for Kyle's  $\lambda$  shocks.

Table C.1: Summary statistics of *ILLIQ*-related variables.

## 6.3 Appendix C tables: robustness and additional evidence

Read Table C.1:

- $\ln ILLIQ$  and  $\ln LSilliq$  are almost perfectly correlated, and both are highly correlated with  $\ln IDVOL$ .
- $DIF$  has a negative average value and is negatively correlated with  $\ln ILLIQ$ ,  $\ln LSilliq$ ,  $\ln |R|$ , and  $\ln IDVOL$ .
- These correlations explain why a raw horse race between *ILLIQ* and *IDVOL* can be hard to interpret.

Read Table C.2:

- The pricing of *ILLIQ*, *DIF*, and *RlnILLIQ* survives controls for Fama-French-Carhart factor loadings.
- The coefficient on *DIF* remains positive and significant in both decomposed specifications.

Read Table C.3:

- Mispricing, measured by *MISP*, is strongly priced, but it does not eliminate the illiquidity premium.
- *DIF* and *RlnILLIQ* remain positive and significant after controlling for mispricing.

Economic message: The key results are not artifacts of conventional risk controls or mispricing proxies.

**Table C.2**  
The effect of *ILLIQ* and components on expected return, controlling for systematic risks

| Explanatory variables                                                                   | (1)            | (2)                 | (3)                 | (4)                 |
|-----------------------------------------------------------------------------------------|----------------|---------------------|---------------------|---------------------|
| $\ln ILLIQ_{j,s-2}$                                                                     | 0.096 (3.16)   |                     |                     |                     |
| $\ln LSIlliq_{j,s-2}$                                                                   |                | 0.108 (3.34)        |                     |                     |
| $\ln  R_{j,s-2} $                                                                       |                |                     | -0.329 (-2.54)      |                     |
| $\ln IDVOL_{j,s-2}$                                                                     |                |                     | 0.096 (3.20)        | 0.092 (3.11)        |
| $DIF_{j,s-2}$                                                                           |                | <b>0.861 (3.83)</b> | <b>0.644 (2.91)</b> |                     |
| $RlnILLIQ_{j,s-2}$                                                                      |                |                     |                     | <b>0.511 (3.79)</b> |
| $IdioVol_{j,s-2}$                                                                       |                |                     |                     | -0.558 (-7.29)      |
| $\beta_{RMrf,j,s-2}$                                                                    | 0.015 (0.17)   | 0.017 (0.20)        | 0.126 (1.86)        | 0.088 (1.23)        |
| $\beta_{SMB,j,s-2}$                                                                     | -0.012 (-0.22) | -0.010 (-0.18)      | 0.043 (1.01)        | 0.048 (1.09)        |
| $\beta_{HML,j,s-2}$                                                                     | 0.100 (2.16)   | 0.096 (2.08)        | 0.083 (2.02)        | 0.100 (2.40)        |
| $\beta_{UMD,j,s-2}$                                                                     | -0.069 (-1.08) | -0.070 (-1.12)      | -0.065 (-1.11)      | -0.083 (-1.40)      |
| Control variables: $Size_{j,s-2}$ , $BM_{j,s-1}$ , $R12lag_{j,s-2}$ , $R1lag_{j,s-1}$ . |                |                     |                     |                     |
| Average Adjusted $R^2$                                                                  | 7.67%          | 7.85%               | 8.65%               | 9.04%               |

The dependent variable in this table is  $(R_j - R_f)$ . This table is similar to Table 1 except that we add to Model (5) the systematic risks,  $\beta_{RMrf}$ ,  $\beta_{SMB}$ ,  $\beta_{HML}$ , and  $\beta_{UMD}$ , the loadings of the respective factors of Fama and French (1993) and Carhart (1997),  $RMrf$ ,  $SMB$ ,  $HML$  and  $UMD$ . They are estimated over a rolling window of past 6 months up to month  $s-2$  and added to the explanatory variables in Model (9).

**Table C.3**  
The effects of illiquidity and mispricing on expected return

| Explanatory variables                                                                   | (1)                   | (2)                   | (3)                   | (4)                   |
|-----------------------------------------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| $MISP_{j,s-2}$                                                                          | <b>-0.017 (-8.73)</b> | <b>-0.017 (-8.75)</b> | <b>-0.016 (-9.51)</b> | <b>-0.015 (-9.15)</b> |
| $\ln ILLIQ_{j,s-2}$                                                                     | 0.106 (2.42)          |                       |                       |                       |
| $\ln LSIlliq_{j,s-2}$                                                                   |                       | 0.114 (2.59)          |                       |                       |
| $\ln  R_{j,s-2} $                                                                       |                       |                       | -0.176 (-0.99)        |                       |
| $\ln IDVOL_{j,s-2}$                                                                     |                       |                       | 0.077 (2.02)          | 0.077 (2.00)          |
| $DIF_{j,s-2}$                                                                           |                       | <b>1.026 (3.89)</b>   | <b>0.832 (3.24)</b>   |                       |
| $RlnILLIQ_{j,s-2}$                                                                      |                       |                       |                       | <b>0.809 (4.58)</b>   |
| $IdioVol_{j,s-2}$                                                                       |                       |                       |                       | -0.602 (-7.47)        |
| Control variables: $Size_{j,s-2}$ , $BM_{j,s-1}$ , $R12lag_{j,s-2}$ , $R1lag_{j,s-1}$ . |                       |                       |                       |                       |
| Average Adjusted $R^2$                                                                  | 5.88%                 | 6.19%                 | 7.68%                 | 7.68%                 |

The dependent variable in this table is  $(R_j - R_f)$ . This table is similar to Table 1 that estimates Model (9) with an added control variable  $MISP_{j,s-2}$  that indicates mispricing. It is constructed by Stambaugh, Yu, and Yuan (2012) by combining each stock's rankings on 11 anomaly variables computed at the end of each month  $s$ . Data are available for the period July 1965 to December 2016 from the authors' web site. The slope coefficients are in percent and the  $t$ -statistics are in parentheses.

Table C.2: Controls for systematic risk loadings.

Table C.3: Illiquidity and mispricing controls.

## 6.4 Appendix C continued: market-shock robustness

Read Table C.4:

- Market illiquidity shocks have substantial variation.
- $dmILLIQ_t$  is highly correlated with  $dmIDVOL_t$ , but  $dmIDVOL_t$  explains only part of the variation in  $dmILLIQ_t$ .

Read Table C.5:

- The negative effect of  $dmILLIQ_t$  on  $RMrf_t$  and  $SMB_t$  is not a January-only phenomenon.
- The interaction  $dmILLIQ_t \times Jan_t$  is insignificant.

Read Table C.6:

- When changes in  $mILLIQ$  and  $mIDVOL$  have opposite signs, returns and benchmark illiquidity measures line up more closely with  $mILLIQ$ .
- $\Delta mILLIQ_t$  has much stronger correlations with benchmark illiquidity variables than  $\Delta mIDVOL_t$ .

Economic message: The time-series evidence consistently favors  $mILLIQ$  over  $mIDVOL$  as a market illiquidity proxy.

**Table C.4**  
Summary statistics of the shocks to the market illiquidity series

| Variables   | Mean(Std.Deviation)(n%) | Pairwise Correlation |           |             |
|-------------|-------------------------|----------------------|-----------|-------------|
| $dmILLIQ_t$ | 0.208 (17.16)           | $dmILLIQ_t$          | $dm R_t $ | $dmIDVOL_t$ |
| $dm R_t $   | -0.017 (16.73)          | 0.437                |           |             |
| $dmIDVOL_t$ | -0.031 (16.40)          | 0.691                | -0.170    |             |
| $dmDIF_t$   | 0.256 (10.51)           | -0.141               | -0.614    | -0.161      |

This table presents summary statistics for the variables  $dmILLIQ_t$ ,  $dm|R_t|$ ,  $dmIDVOL_t$ , and  $dmDIF_t$  for the period January 1955-December 2016, described in Table 2.

**Table C.5**  
The effect of market illiquidity shocks on realized stock returns, controlling for the January effect

| Explanatory variables | Dependent variable |                |
|-----------------------|--------------------|----------------|
|                       | $RMrf_t$           | $SMB_t$        |
| $dmILLIQ_t$           | -0.120 (-11.57)    | -0.051 (-7.68) |
| $Jan_t$               | -0.140 (-0.24)     | 1.650 (4.09)   |
| $dmILLIQ_t * Jan_t$   | -0.029 (-0.94)     | -0.001 (-0.05) |
| $RMrf_t$              |                    | 0.087 (2.80)   |
| Adjusted $R^2$        | 0.23               | 0.17           |

The variables are defined in Table 2.  $Jan_t = 1$  in the month of January and zero otherwise. The time-series regressions include intercepts (not reported). The slope coefficients are in percent. The  $t$ -statistics are presented in parentheses, employing the robust estimation of standard errors by White (1980).

Table C.4: Summary statistics for market illiquidity shocks.

Table C.5: January-effect robustness.

**Table C.6**  
Opposite changes in  $mILLIQ$  and  $mIDVOL$ : their effects on returns and their relation to changes in benchmark market illiquidity variables

|     |                                     | A. Means of variables                                   |                                                         | B. Correlations                      |                                      |
|-----|-------------------------------------|---------------------------------------------------------|---------------------------------------------------------|--------------------------------------|--------------------------------------|
|     |                                     | (1)<br>$\Delta mILLIQ_t > 0$ &<br>$\Delta mIDVOL_t < 0$ | (2)<br>$\Delta mILLIQ_t < 0$ &<br>$\Delta mIDVOL_t > 0$ | (3)<br>... with<br>$\Delta mILLIQ_t$ | (4)<br>... with<br>$\Delta mIDVOL_t$ |
| (1) | N                                   | 75                                                      | 91                                                      | 804                                  |                                      |
| (2) | $RMrf_t$                            | -2.585 (-4.57)                                          | 1.274 (2.94)                                            | -0.500                               | -0.235                               |
| (3) | $SMB_t$                             | -0.903 (-3.02)                                          | 0.281 (1.17)                                            | -0.350                               | -0.184                               |
| (4) | $\Delta m\lambda_t$<br>(n = 27, 38) | 5.881 (1.84)                                            | -5.237 (-1.90)                                          | 0.326                                | 0.040                                |
| (5) | $\Delta mQSP_t$<br>(n = 38, 42)     | 4.952 (1.46)                                            | -6.284 (-2.02)                                          | 0.315                                | 0.066                                |
| (6) | $\Delta mESP_t$                     | 13.387 (7.47)                                           | -9.579 (-6.83)                                          | 0.474                                | -0.055                               |
| (7) | $iPSIlliq_t$<br>(n = 57, 71)        | 2.149 (2.01)                                            | -1.720 (-3.17)                                          | 0.307                                | 0.069                                |

Panel A presents the means of variables for two subsamples of months in which  $\Delta mILLIQ_t$  and  $\Delta mIDVOL_t$  have opposite signs, where  $\Delta$  indicates the first differences of the series that are presented in Table 2. In addition to the variables that are described in Table 2, we include the following variables:  $mQSP_t$ , the logarithm of the value-weighted market average of the quoted relative bid-ask spread, the dollar spread divided by the spread midpoint, using CRSP daily data for NYSE\AMEX stocks;  $mESP_t$ , the logarithm of the market average of the effective relative bid-ask spread, calculated by Abdi and Ronaldo (2017);  $iPSIlliq_t$ , the innovations in the market liquidity series of Pastor and Stambaugh (2003) multiplied by -1 to make it an illiquidity series. The sample period is January 1950-December 2016, 804 months. The series  $m\lambda_t$  is available for 1983-2009 (324 months), the series  $\Delta mQSP_t$  is available for 1993-2016 (288 months), and the sample period for  $iPSIlliq_t$  is August 1962-December 2016 (653 months). N is the default number of months in each estimation and n is the sample size for a particular variable with a shorter sample period. The numbers in parentheses are  $t$ -statistics. The numbers in panel A are in percent. Panel B presents the pair-wise correlations of  $\Delta mILLIQ_t$  and  $\Delta mIDVOL_t$  with the other variables.

Table C.6: Opposite changes in  $mILLIQ$  and  $mIDVOL$ .

## 6.5 Figure C.1: Market illiquidity during the 1987 and 2008 crises

Read:

- Figure C.1 compares market  $ILLIQ$  and market  $IDVOL$  around the October 1987 crash and the October 2008 crisis.
- In both crises, market  $ILLIQ$  rises sharply, in line with benchmark illiquidity measures.
- Market  $IDVOL$  moves much less and can even fail to capture the large crisis-time spike in illiquidity.

Economic message: In major liquidity crises,  $mILLIQ$  behaves like a market illiquidity measure, while  $mIDVOL$  misses important crisis dynamics.

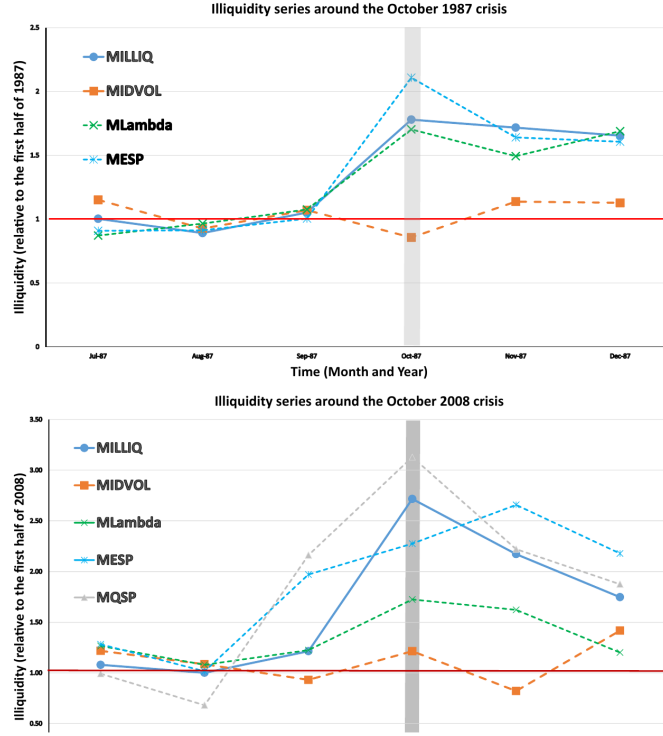


Figure C.1: Market *ILLIQ* and *MIDVOL* during the 1987 and 2008 crises.

## 7 Short summary

- Amihud and Noh revisit whether Amihud's *ILLIQ* measure is priced because of illiquidity or merely because it contains inverse dollar volume.
- Lou and Shu's decomposition misses the covariance between absolute returns and inverse dollar volume.
- The missing component, *DIF*, is positively and significantly priced in the cross-section.
- The residual part of  $\ln ILLIQ$  not explained by  $\ln IDVOL$  is also positively priced.
- The results survive controls for size, book-to-market, reversal, momentum, idiosyncratic volatility, systematic risk loadings, mispricing, sentiment, and seasonality.
- In time-series tests, aggregate market *ILLIQ* shocks explain realized stock returns better than aggregate market *IDVOL* shocks.
- The sign of the  $dmILLIQ$  coefficient is consistent with theory: unexpected increases in illiquidity lower contemporaneous realized returns.
- Benchmark microstructure measures and crisis episodes support *mILLIQ* as a broader illiquidity proxy than *mIDVOL*.

## 8 Conclusion

### 8.1 Core conclusion

The paper concludes that Amihud’s *ILLIQ* contains priced illiquidity information beyond inverse dollar volume. Trading volume is related to liquidity, but the return-impact component embedded in *ILLIQ* is economically meaningful and cannot be discarded.

### 8.2 Economic intuition

Illiquidity is about the price movement associated with trading, not only the amount traded. *IDVOL* captures scarcity of trading activity, but *ILLIQ* also captures how much prices move per dollar traded. The omitted covariance component measures part of this relation and is therefore relevant for expected returns.

### 8.3 Incremental contribution revisited

The paper turns a critique of *ILLIQ* into a sharper understanding of what the measure contains. It shows that the part omitted by the Lou-Shu decomposition is both theoretically related to illiquidity and empirically priced.

### 8.4 Implications

Empirical asset-pricing tests should not treat inverse dollar volume as a complete substitute for Amihud illiquidity. For cross-sectional expected returns and aggregate market illiquidity shocks, *ILLIQ* captures broader liquidity information.

### 8.5 Limitations

The paper works with proxy measures and reduced-form return regressions. *ILLIQ* does not capture every dimension of liquidity, and no single low-frequency measure can fully summarize trading costs. The results nevertheless show that *ILLIQ* remains empirically useful.

### 8.6 Takeaway

The enduring takeaway is that price impact matters. Amihud’s *ILLIQ* is not merely inverse volume in disguise; it embeds additional illiquidity information that helps explain both expected stock returns and realized market returns during illiquidity shocks.